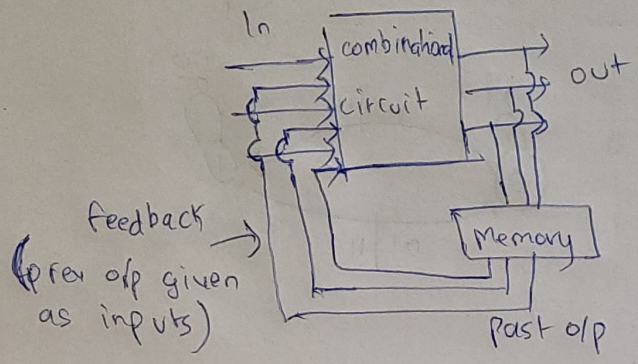
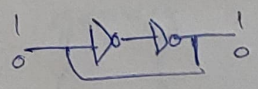
UNIT -3

NOTES

In sequential circuits, present o/p depends on present i/p as well as past o/p.

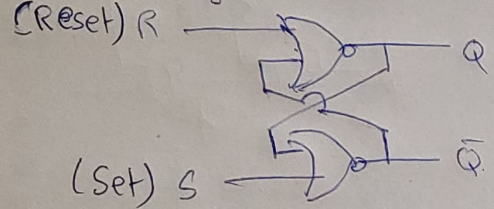




Memory (early version)

SR Latch (using NOR)

Basic storage element → latch (as it latches 0 or 1)



- Reset means $Q = 0$
- Set means $Q = 1$

nand gate

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	0

if any i/p is '1', o/p is '0'

case 1, $S=0, R=1$ makes $Q=0$ (as $R=1$)

$\Rightarrow \bar{Q} = 1$

(i/p removed) $S=0, R=0$ (memory) $\bar{Q}$ was 1.

$\therefore Q = 0$ as $R=0$ & $\bar{Q}=1$ $\bar{Q}=1$ is also verified (0,0)

case 2, $S=1, R=0$ makes $\bar{Q}=0$ (as $S=1$)

$\Rightarrow Q = 1$

(i/p removed) $S=0, R=0$ (memory) Q was 1

$\therefore \bar{Q} = 0$ as $S=0$ & $Q=1$

$Q=1$ is also verified (0,0)

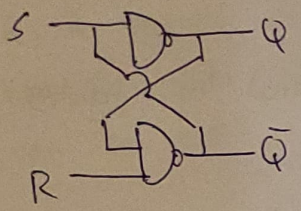
case 3, $S=1, R=1$ gives $Q=0$ & $\bar{Q}=0$

This is invalid as $Q = \bar{Q}$

$\therefore$ This case is not used/ignored

S	R	Q	$\bar{Q}$
0	0	memory	
0	1	0	1 ← set
1	0	1	0 ← reset)
1	1	not used ($Q=0$ $\bar{Q}=0$)	

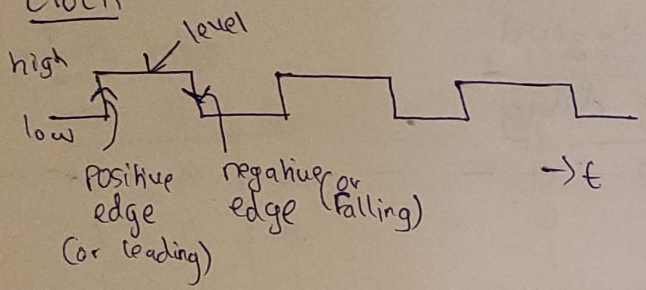
Nand SR-Latch



S	R	Q	$\bar{Q}$	
0	0	not used (gives $Q=1, \bar{Q}=1$)		
0	1	1	0	← set
1	0	0	1	← reset
1	1	memory		

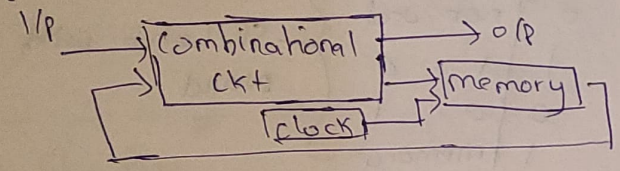
if even one i/p to nand gate is 0, o/p is forced to 1

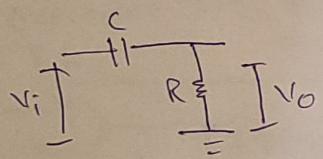
Clock



FF
- edge triggered: works during edges
- level triggered: FF works during levels
- Duty cycle: $\frac{\text{time for which signal is high}}{\text{total time}}$

$= \frac{t/2}{t} = 50\% = \frac{1}{2}$





$v_i - v_C - v_0 = 0$

$v_i = v_C + v_0$

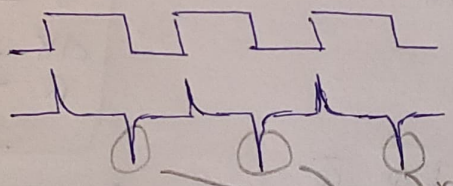
$= \frac{Q}{C} + iR$

$= \frac{\int i\,dt}{C} + iR$ (iR: very small)

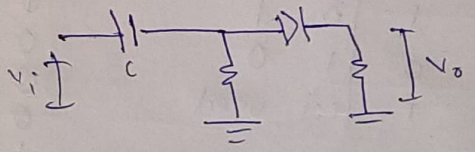
$= \frac{\int i\,dt}{C} = \frac{1}{C}\int \frac{v_0}{R}\,dt$

$\frac{dv_i}{dt} = \frac{1}{RC} v_0$

$\Rightarrow v_0 = RC \frac{dv_i}{dt}$



pos/negative edge can be removed by diode.



Latch vs Flip flop

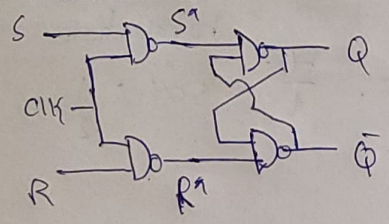
Latch

- Only feedback path
- Async in nature (o/p is obtained as soon as i/p is given)
- No clock
- level triggered
- No delay

FF

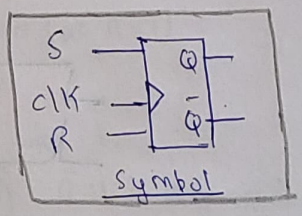
- feedback path + external logic gates
- Sync in nature (o/p is obtained when i/p & clk pulse are given)
- clock is needed
- Edge triggered
- Has a delay

SR Flip Flop



- $S' = \bar{S} + \overline{clk}$
- $R' = \bar{R} + \overline{clk}$

S'	R'	Q	$\bar{Q}$	
0	0	not used		SR Latch with Nand
0	1	1	0	
1	0	0	1	
1	1	memory		



- Truth table

clk	S	R	Q	$\bar{Q}$
0	x	x	memory	
1	0	0	memory	
1	0	1	0	1
1	1	0	1	0
1	1	1	not used	

OR

clk	S	R	Q_{n+1}
0	x	x	Q_n
1	0	0	Q_n
1	0	1	0
1	1	0	1
1	1	1	not used

Q_{n+1} → next State
Q_n → Present State (stored in FF)

- characteristic table (clk=1)

	Q_n	S	R	Q_{n+1}
1)	0	0	0	0
1)	0	0	1	0
2)	0	1	0	1
	0	1	1	not used(x)
4)	1	0	0	1
3)	1	0	1	0
4)	1	1	0	1
	1	1	1	not used(x)

- excitation table

	Q_n	Q_{n+1}	S	R
1)	0	0	0	x
2)	0	1	1	0
3)	1	0	0	1
4)	1	1	x	0

↑ can do Kmap for S or R using this

Q_n \ SR	00	01	11	10
0	0	0	x	1
1	1	0	x	1

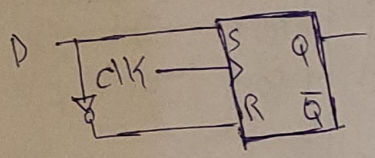
$$Q_{n+1} = S + Q_n\bar{R}$$

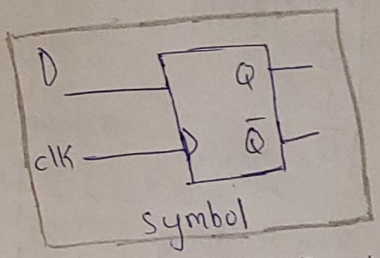
from excitation table,

$$S = Q_{n+1} \qquad R = \bar{Q}_{n+1}$$

D Flip Flop (data)

in SR, 2 inputs are needed to manipulate the FF, for changing data stored, S and R are always complements.
∴ D can be used





- Truth table

clk	D	Q_{n+1}
0	X	Q_n
1	0	0
1	1	1

- Characteristic table (clk=1)

Q_n	D	Q_{n+1}
0	0	0
0	1	1
1	0	0
1	1	1

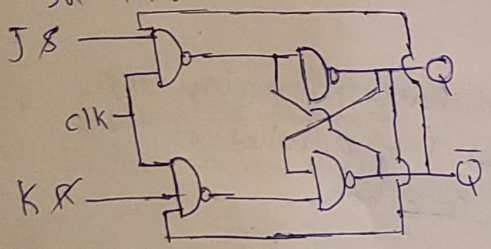
- Q_{n+1} follows D
- $(Q_{n+1} = D)$

- Excitation table

Q_n	Q_{n+1}	D
0	0	0
0	1	1
1	0	0
1	1	1

JK Flip Flop

→ to overcome the invalid condition that occurs when S=1 & R=1 in SR FF.



- clk = 0, memory
- clk = 1, J=1, K=0 Q=1, $\bar{Q}$=0
- clk = 1, J=0, K=1 Q=0, $\bar{Q}$=1
- clk = 1, J=0, K=0 memory

} same as SR FF

- clk = 1, J=1, K=1

assume Q=0 & $\bar{Q}$=1

For K gate, i/p is 1,1,0 → 1
J gate, i/p is 1,1,1 → 0
this gives,

- Q=1 & $\bar{Q}$=0

⇒ K gate gives 0 (1,1,1)
⇒ J gate gives 1 (1,1,0)
this gives
Q=0, $\bar{Q}$=1

Q & $\bar{Q}$ are constantly fluctuating blw 0&1

Q → 0,1,0,1,0 . . . . .
$\bar{Q}$ → 1,0,1,0,1 . . . .

Q & $\bar{Q}$ are said to be 'racing'

∴ not used changes to $\overline{Q_n}$ $(Q_{n+1} = \overline{Q_n})$

• Truth table:

clk	J	K	Q_{n+1}
0	x	x	memory (Q_n)
1	0	0	memory (Q_n)
1	0	1	0
1	1	0	1
1	1	1	$\bar{Q}_n$ (toggle)

Q_{n+1} is racing 0,1,0,1.....

• Characteristic table (clk=1)

	Q_n	J	K	Q_{n+1}
1)	0	0	0	0 (Q_n)
1)	0	0	1	0
2)	0	1	0	1
2)	0	1	1	1 ($\bar{Q}_n$)
4)	1	0	0	1 (Q_n)
3)	1	0	1	0
4)	1	1	0	1
3)	1	1	1	0 ($\bar{Q}_n$)

• Excitation table

	Q_n	Q_{n+1}	J	K
1)	0	0	0	x
2)	0	1	1	x
3)	1	0	x	1
4)	1	1	x	0

•

Q_n \ Q_{n+1}	0	1
0	0	1
1	x	x

for J:

• $J = Q_{n+1}$

•

Q_n \ Q_{n+1}	0	1
0	x	x
1	1	0

for K:

• $K = \bar{Q}_{n+1}$

•

Q_n \ JK	00	01	11	10
0	0	0	1	1
1	1	0	0	1

redundant

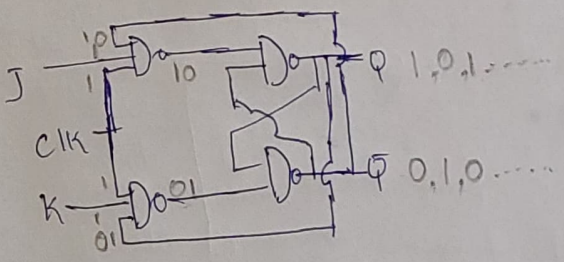
for Q_{n+1}:

• $Q_{n+1} = Q_n\bar{K} + \bar{Q}_nJ$

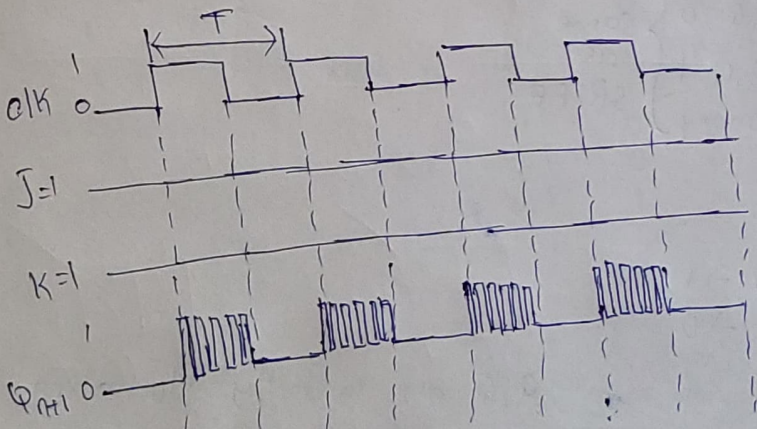
• Race Around Condition

When J=1, K=1, clk=1,

Q=1 & $\bar{Q}$=0 (assume)



← output continuously changing from 0 to 1 and 1 to 0. Called as race around condition.



race around occurs only when J=1, K=1 & clk=1

As FF have delay in giving output, T/2 must be bigger than the delay.

Conditions to overcome delay:

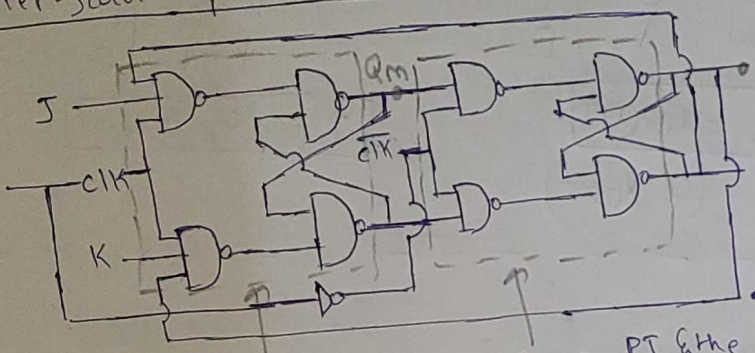
i) T/2 < delay of flip flop

ii) Edge triggering (doesn't allow enough time for racing)

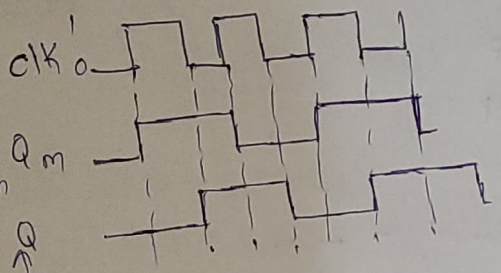
iii) Master-slave

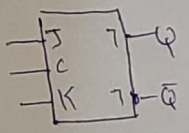
- Racing → uncontrolled
- toggling → controlled racing
- Master slave - operation is same as negative edge triggered.

Master-Slave Operation of JK flip flop



- master → positive level triggered
- slave → negative level triggered
- ∴ Master responds to its J & K inputs before slave. If J=1 & K=0, master sets on positive transition of clock. The high Q output of master drives J input of slave, and on -ve transition, the slave sets.
- Ill'y J=0, K=1 resets master on PT & the high $\bar{Q}$ makes slave reset on NT.
- If J=1, K=1, master toggles on PT & then slave toggles on NT.





When clk=1, (PT) master operates and slaves doesn't, i.e. slave shows memory / prev state

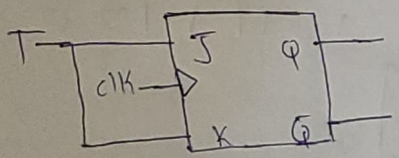
When clk=0, (NT) master doesn't operate, i.e. it shows memory and slave operates.

∴ The racing has been avoided as the o/p only changes state once now, called as toggling.

clk	J	K	Q_{n+1}
1	1	1	$\bar{Q}_n$

- Master is transparent while clock is high.
- The "1" means postponed output. The contents of master are shifted into slave on NT & Q changes state.

T Flip flop (Toggle)



Truth table

clk	T	Q_{n+1}
0	X	Q_n (memory)
1	0	Q_n (memory)
1	1	$\bar{Q}_n$ (Toggle)

used when only toggling action of JK flip flop is required.

Note:

- PT - positive transition of clk
- NT - negative "
- transparent - Any changes done to input is reflected immediately in the output.

• Truth table:

Clk	T	Q_{n+1}
0	X	Q_n
1	0	Q_n
1	1	$\overline{Q_n}$

• characteristic table:

Q_n	T	Q_{n+1}
0	0	0
0	1	1
1	0	1
1	1	0

$Q_{n+1} = Q_n \oplus T$

(Q_n XOR T)

• Excitation table

Q_n	Q_{n+1}	T
0	0	0
0	1	1
1	0	1
1	1	0

$T = Q_n \oplus Q_{n+1}$

• Flip Flop Conversion

1) Identify available & required flip flop
2) Make characteristic table for required FF
3) Make excitation table for available FF
4) write boolean expression for available FF
5) Draw the circuit

• JK to D FF

1) available = JK
   required = D

2)

Q_n	D	Q_{n+1}
0	0	0
0	1	1
1	0	0
1	1	1

characteristic table for D

3)

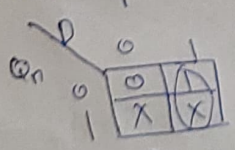
Q_n	Q_{n+1}	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

excitation table for JK

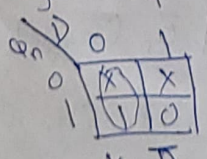
• Combine both

Q_n	D	Q_{n+1}	J	K
0	0	0	0	X
0	1	1	1	X
1	0	0	X	1
1	1	1	X	0

4) Get exp. for J & K using K-Map using this table



J = D

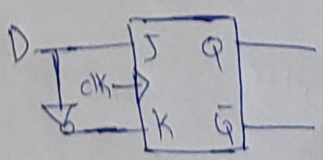


K = $\overline{D}$

5) First draw available FF

now,

J = D and K = $\overline{D}$



• T to D FF

required = D
available = T

Q_n	D	Q_{n+1}
0	0	0
0	1	1
1	0	0
1	1	1

characteristic table

Q_n	Q_{n+1}	T
0	0	0
0	1	1
1	0	1
1	1	0

excitation table

combining

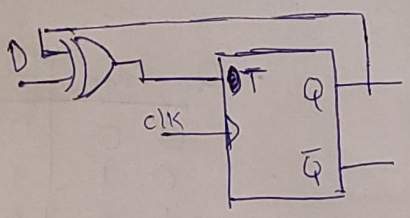
Q_n	D	Q_{n+1}	T
0	0	0	0
0	1	1	1
1	0	0	1
1	1	1	0

now get T

Q_n \ D	0	1
0	0	1
1	1	0

$\therefore T = \bar{Q}_n D + Q_n \bar{D}$

$T = Q_n \oplus D$



• SR to JK FF

available = SR
required = JK

Q_n	J	K	Q_{n+1}
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	0

characteristic table

Q_n \ JK	00	01	11	10
0	0	0	1	1
1	X	0	0	X

dont use

$S = \bar{Q}_n J$

Q_n	Q_{n+1}	S	R
0	0	0	X
0	1	1	0
1	0	0	1
1	1	X	0

excitation table

combining

Q_n	J	K	Q_{n+1}	S	R
0	0	0	0	0	X
0	0	1	0	0	X
0	1	0	1	1	0
0	1	1	1	1	0
1	0	0	1	X	0
1	0	1	0	0	1
1	1	0	1	X	0
1	1	1	0	0	1

now, get S & R

Q_n \ JK	00	01	11	10
0	X	X	0	0
1	0	1	1	0

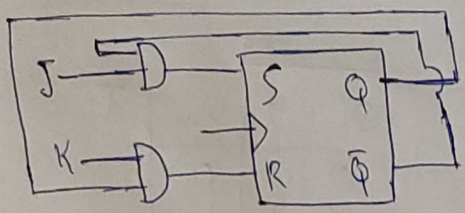
dont use

$R = Q_n K$

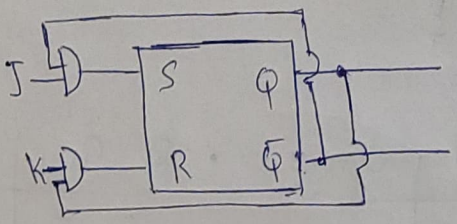
Dont use
Dont care

$S = \bar{Q}_n J$ [redacted]

$R =$ [redacted] $+ Q_n K$



≈



• SR to T FF

available = SR
required = T

• characteristic table

Q_n	T	Q_{n+1}
0	0	0
0	1	1
1	0	1
1	1	0

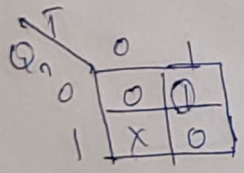
• excitation table

Q_n	Q_{n+1}	S	R
0	0	0	X
0	1	1	0
1	0	0	1
1	1	X	0

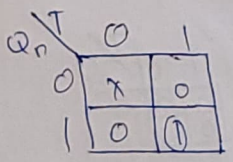
combine

• derive S & R using K-Map

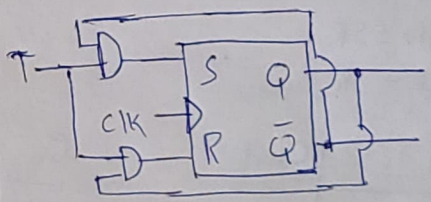
Q_n	T	Q_{n+1}	S	R
0	0	0	0	X
0	1	1	1	0
1	0	1	X	0
1	1	0	0	1



$S = \bar{Q}_n T$



$R = Q_n T$



extras: 1) JK to SR ($J=S$ & $K=R$)
2) T to SR ($T = \bar{Q}S + QR$)

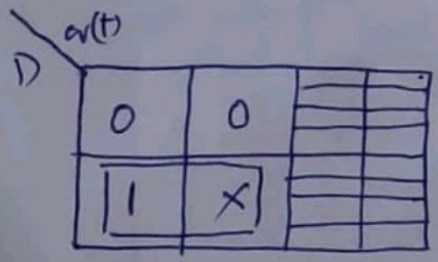
• use dont care to form extra groups only in case of:
D to SR, T to SR,

Flip-flop Conversion

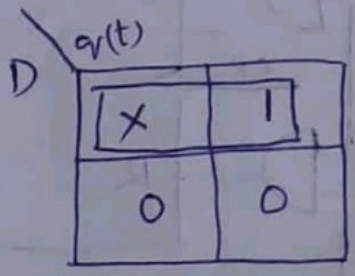
i) SR flip-flop to D flip-flop

Conversion Table

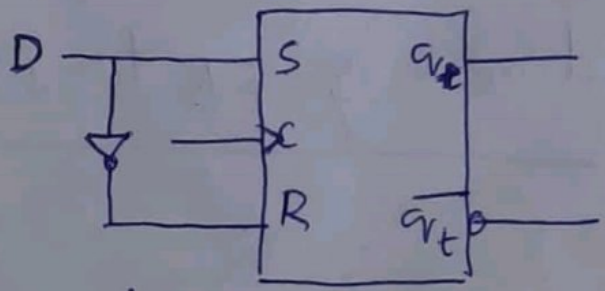
D I/p	outputs		SR I/p	
D	q_t	q_{t+1}	S	R
0	0	0	0	X
1	0	1	1	0
0	1	0	0	1
1	1	1	X	0



$S = D$



$R = \overline{D}$

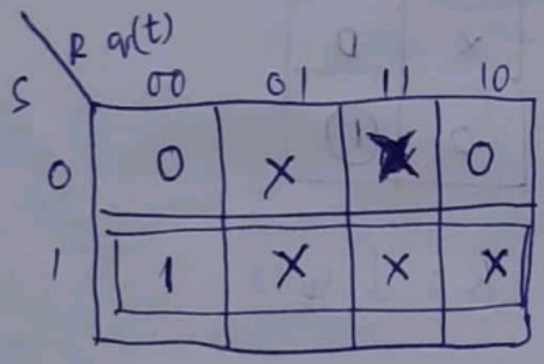


logical diagram of SR to D Conversion FF

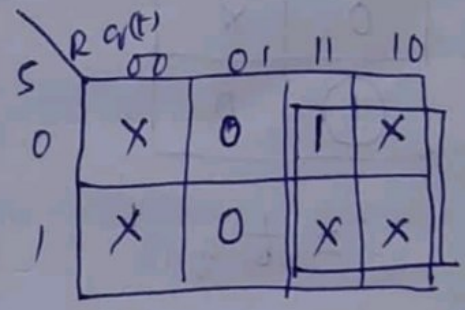
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16(v) JK to SR flip-flop

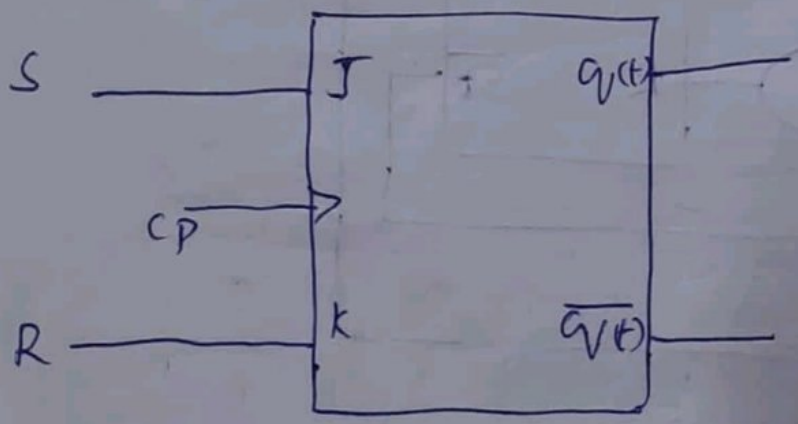
S	R	$q_n(t)$	$q_n(t+1)$	J	K
0	X	0	0	0	X
1	0	0	1	1	X
0	1	1	0	X	1
X	0	1	1	X	0



$J = S$



$K = R$

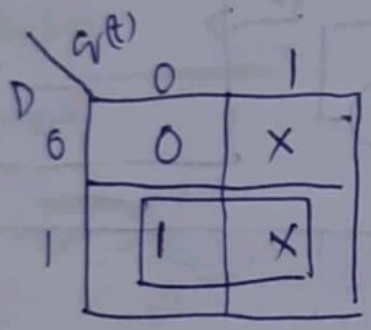


logical diagram of JK to SR

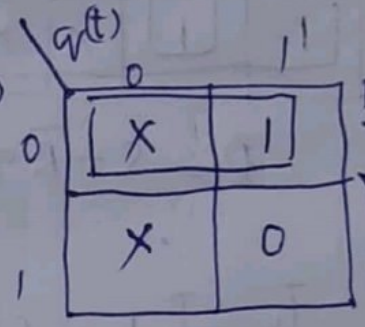
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v) JK to D flip-flop

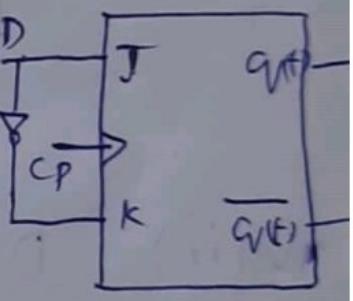
D	$q_n(t)$	$q_n(t+1)$	J	K
0	0	0	0	X
1	0	1	1	X
0	1	0	X	1
1	1	1	X	0



$J = D$

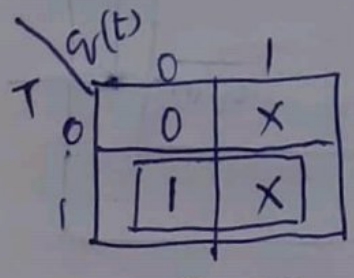


$K = \overline{D}$

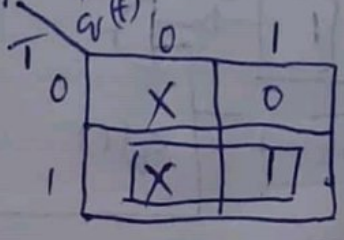


vi) JK to T flip-flop

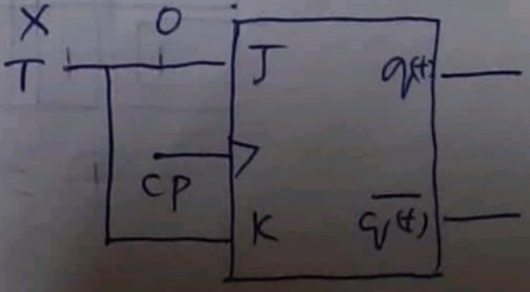
T	$q_n(t)$	$q_n(t+1)$	J	K
0	0	0	0	X
1	0	1	1	X
1	1	0	X	1
0	1	1	X	0



$J = T$



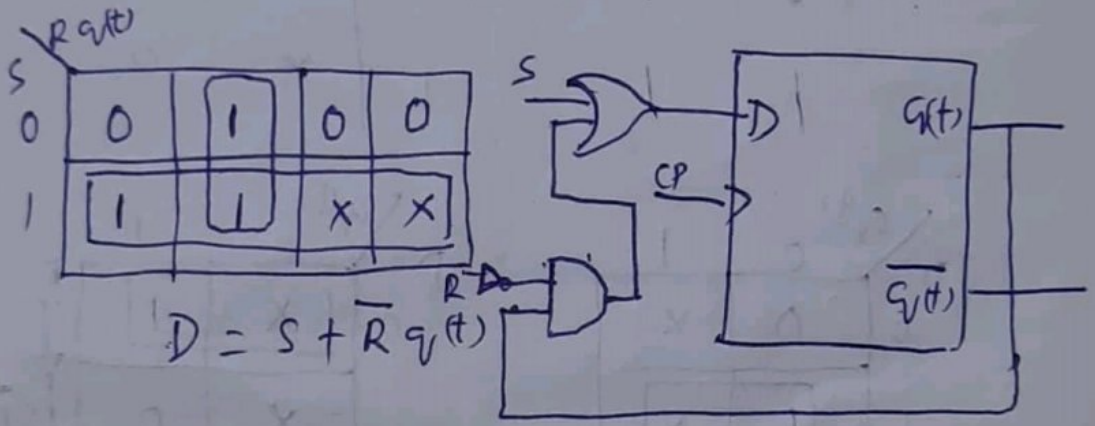
$K = T$



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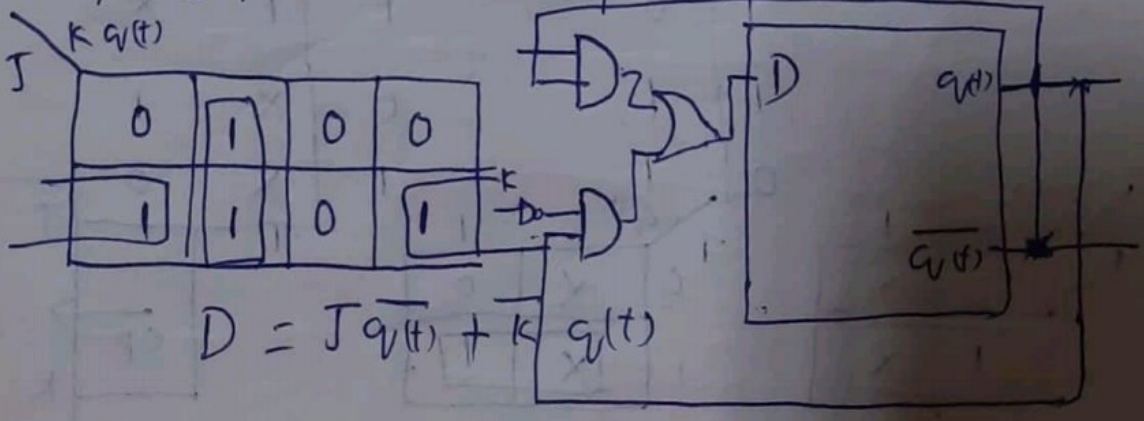
vii) D to SR flip-flop

S	R	$q_v(t)$	$q_v(t+1)$	D
0	X	0	0	0
1	0	0	1	1
0	1	1	0	0
X	0	1	1	1



viii) D to JK flip-flop

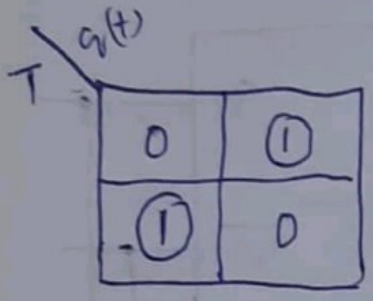
J	K	$q_v(t)$	$q_v(t+1)$	D
0	X	0	0	0
1	X	0	1	1
X	1	1	0	0
X	0	1	1	1



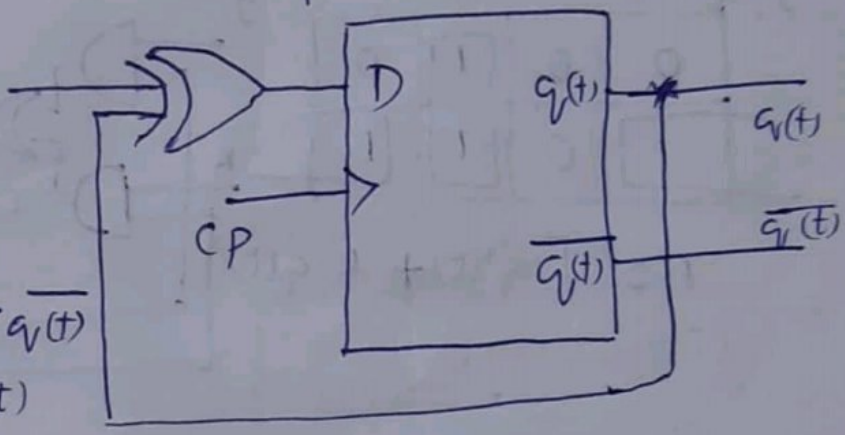
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ix) D flip-flop to T flip-flop

T	$q_n(t)$	$q_n(t+1)$	D
0	0	0	0
1	0	1	1
1	1	0	0
0	1	1	1

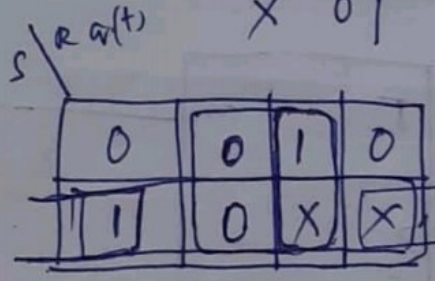


$$D = \bar{T}\, q_n(t) + T\, \overline{q_n(t)}$$
$$= T \oplus q_n(t)$$

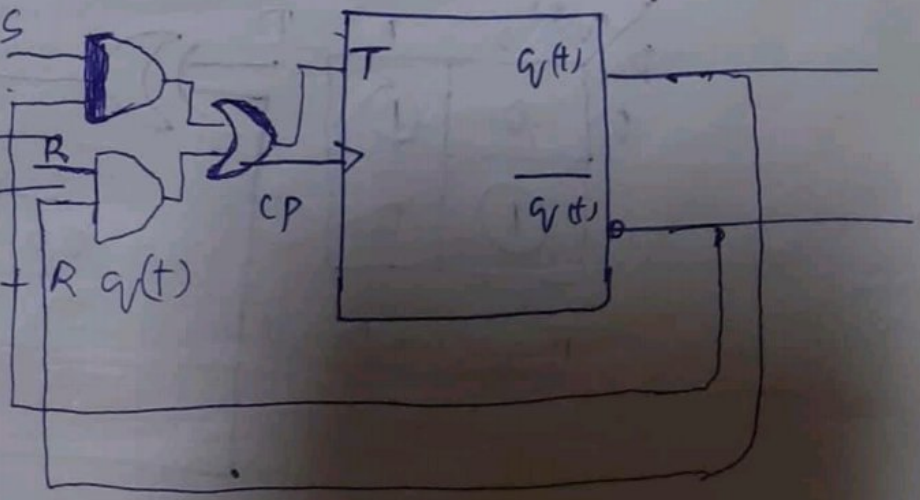


x) To Convert from T to SR flip-flop.

S	R	$q_n(t)$	$q_n(t+1)$	T
0	X	0	0	0
1	0	0	1	1
0	1	1	0	1
X	0	1	1	0



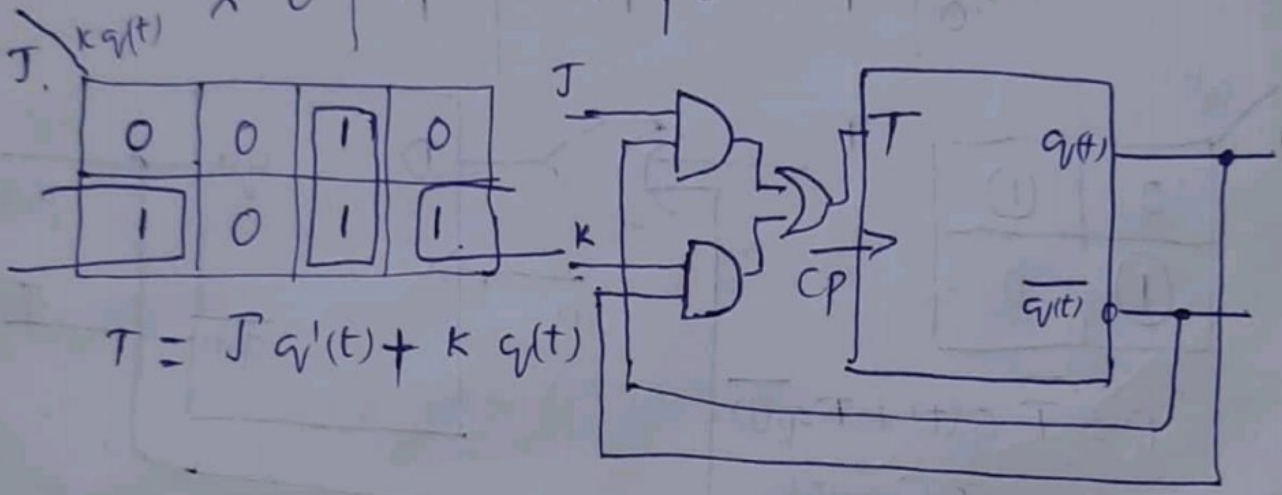
$$T = S\, q'(t) + R\, q(t)$$



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xi) T to JK flip-flop

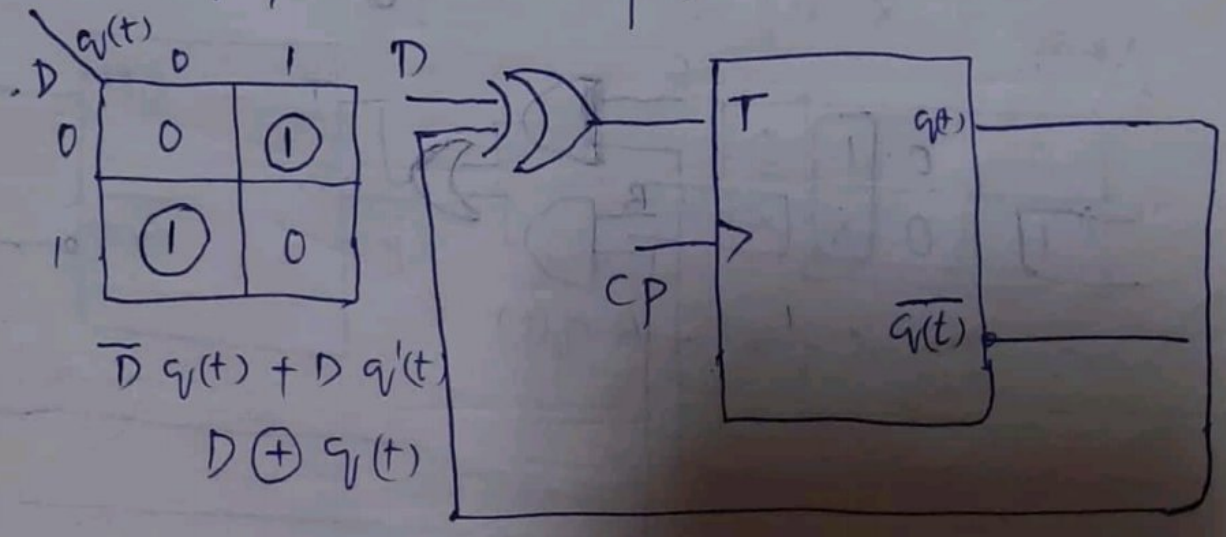
J	K	$q(t)$	$q(t+1)$	T
0	x	0	0	0
1	x	0	1	1
x	1	1	0	1
x	0	1	1	0



$T = J q'(t) + K q(t)$

xii) T to D flip-flop

D	$q(t)$	$q(t+1)$	T
0	0	0	0
1	0	1	1
0	1	0	1
1	1	1	0



$\bar{D}\, q(t) + D\, q'(t)$

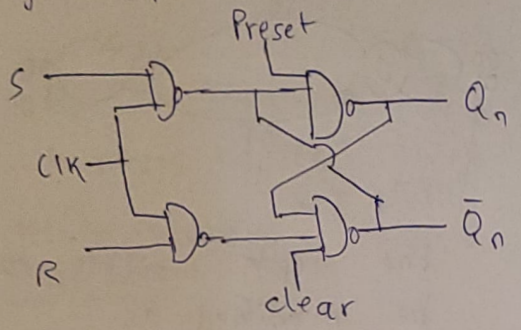
$D \oplus q(t)$

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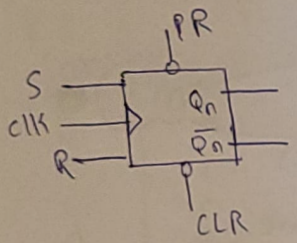
• Preset & clear Inputs

→ They are direct inputs / overriding inputs / async inputs

→ Sync inputs are S, R, J, K, D & T



- If Preset = 0, Q_n is 1 no matter other conditions
- If clear = 0, Q_n = 0 as $\bar{Q}_n$ = 1 no matter other conditions



PR-Preset
CLR-clear
bubble as they are active low inputs (when it is low, output is 1)

Preset	Clear	Q_n
0	0	not used
0	1	1
1	0	0
1	1	FF operates normally

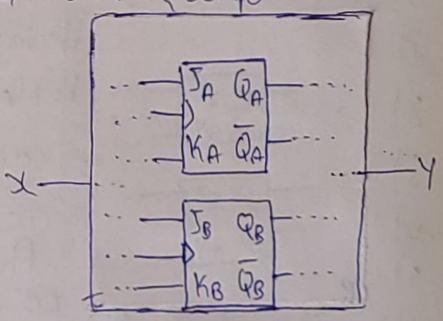
• State diagram

State table: tells relation blw present state, next state & output

(internal clk is not known, sample table:)

Present state		x (input)	Next state		y (output)
Q_A	Q_B		Q_A^+	Q_B^+	
0	0	1	1	0	1

- Draw arrows blw the states using the state table.

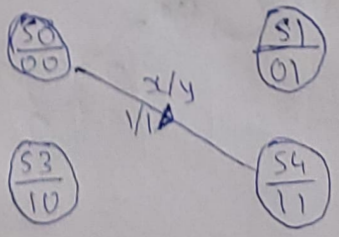


Sequential circuit

The various states are:

- S_0 = Q_A 0, Q_B 0
- S_1 = 0 1
- S_2 = 1 0
- S_3 = 1 1

each of these states is shown by a bubble



- For JK flip flop

state table:

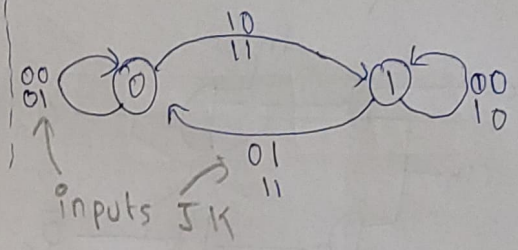
Present	I/P		next state
Q_n	J	K	Q_{n+1}
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	0

The states are:

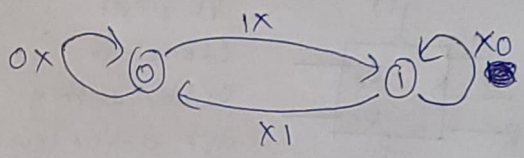
$S_0 = 0$

$S_1 = 1$

(present state)



the inputs written can be modified as:



x → dont care

State equation:

LHS = RHS

LHS → next state = 1

RHS → present state & input

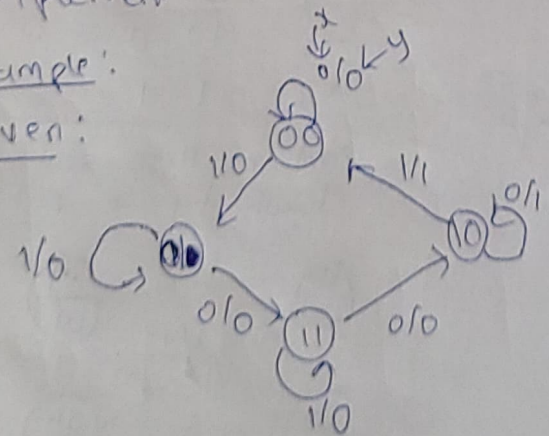
- $Q_{n+1} = \bar{Q}_n J + Q_n \bar{K}$ (for JK)

Design Procedure:

1) State diagram / timing diagram of circuit to be designed is given
2) Obtain state table using ↑
3) Number of states can be reduced by state reduction method
4) Do state assignment
5) Find number of flip flops needed & assign letter symbols
6) Decide type of FF to be used
7) Find excitation table of circuit from state table
8) Get expression for circuit o/p & FF i/p
9) Implement the circuit

example:

Given:



{ x → input
P.S - Present state N.S - next state

Answer:

P.S		N.S				(output) y	
		x = 0		x = 1			
Q_A	Q_B	Q_A^+	Q_B^+	Q_A^+	Q_B^+	x = 0	x = 1
0	0	0	0	0	1	0	0
0	1	1	1	0	1	0	0
1	0	1	0	0	0	1	1
1	1	1	0	1	1	0	0

State reduction:

NOTE

State assignment: assigning binary numbers to the states.

ex: If there are two states: a & b can be represented as

a=0 & b=1

but if there are 4 states: a, b, c & d, we need 2 binary numbers for each state: ex: a=00 b=01 c=10 d=11

ex: (for reduction)

State table

P.S	N.S x=0	N.S x=1	Output x=0	Output x=1
a	a	b	0	0
b	c	d	0	0
c	a	d	0	0
d	e=g	f	0	1
e	a	f	0	1
~~f~~	~~g~~	~~f~~	~~0~~	~~1~~
~~g~~	~~a~~	~~f~~	~~0~~	~~1~~

If N.S & output are same, the state can be reduced

In given table,

e=g ∴ either e or g can be removed from table & diagram

now e=g is proved. using this, d & f become same.

∴ d=f

Continued:

- In the state table formed using the given diagram, state reduction cannot occur (no duplicates).
- state assignment is not needed as 00, 10, 11, 01 are already given to states. If in question letters like a, b, c, d are given, then assignment must be done.
- We need 2 FF (4 states) let them be A (Q_A) and B (Q_B) (T-flip flop)
- Circuit excitation table: (using state table)

P.S. Q_A	P.S. Q_B	x	N.S Q_A^+	N.S Q_B^+	FF i/p T_A	FF i/p T_B	y
0	0	0	0	0	0	0	0
0	0	1	0	1	0	1	0
0	1	0	1	1	1	0	0
0	1	1	0	1	0	0	0
1	0	0	1	0	0	0	1
1	0	1	0	0	1	0	1
1	1	0	1	0	0	1	0
1	1	1	1	1	0	0	0

(compare P.S & N.S)

- The FF i/p column is filled using truth table of the FF chosen

clk	T	Q_{n+1}
0	x	Q_n
1	0	Q_n
1	1	$\overline{Q_n}$

Q_n	Q_{n+1}	T
0	0	0
0	1	1
1	0	1
1	1	0

} using this

- Use K-map to get the expression for FF. i/p (T_A & T_B) & y

Q_A \ $Q_B x$	00	01	11	10
0	0	0	0	1
1	0	1	0	0

$\therefore T_A = \bar{Q}_A Q_B \bar{x} + Q_A \bar{Q}_B x$

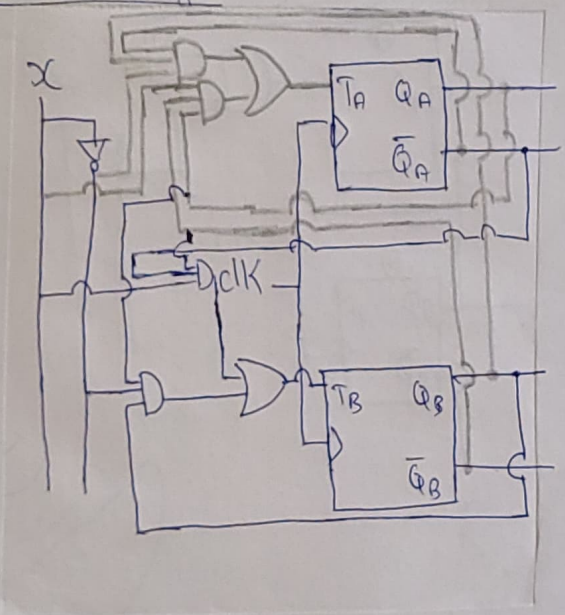
Q_A \ $Q_B x$	00	01	11	10
0	0	0	0	0
1	0	0	0	1

$T_B = \bar{Q}_A \bar{Q}_B x + Q_A Q_B \bar{x}$

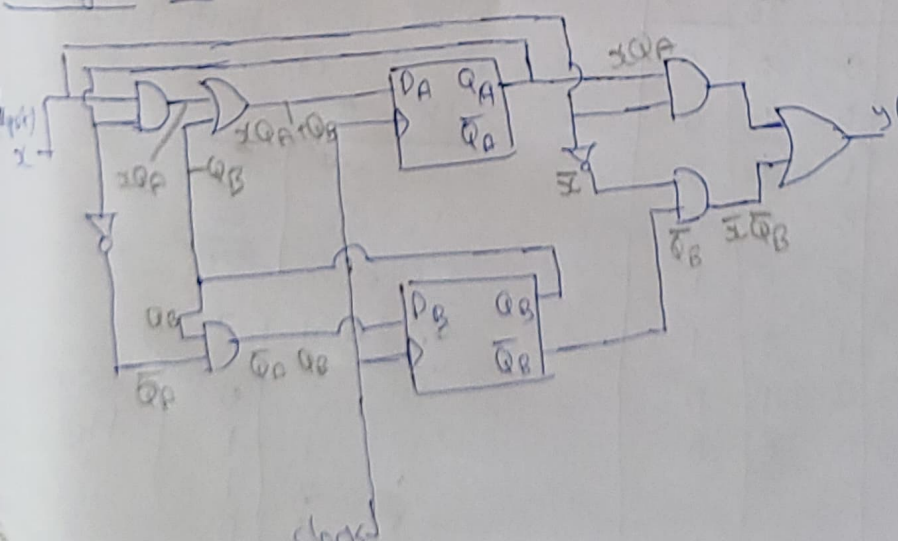
Q_A \ $Q_B x$	00	01	11	10
0	0	0	0	0
1	1	1	0	0

$y = Q_A \bar{Q}_B$

- Circuit design:



Analysis of Clocked Sequential Circuits



Step 1:
Find i/p & o/p equations

$D_A = x\bar{Q}_A + Q_B$

$D_B = \bar{Q}_A Q_B$

$y = \bar{x}\bar{Q}_B + x\bar{Q}_A$

Step 2: State table (find N.S & y)

$D = Q_{n+1}$

P.S Q_A	P.S Q_B	x	N.S Q_A^+	N.S Q_B^+	y
0	0	0	0	0	1
0	0	1	0	0	0
0	1	0	1	1	0
0	1	1	1	1	0
1	0	0	0	0	1
1	0	1	1	0	1
1	1	0	1	0	0
1	1	1	1	0	1

$\therefore D_A = Q_A^+$

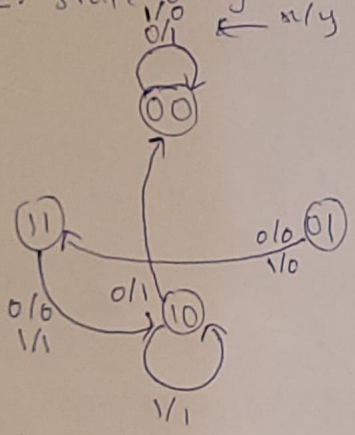
$\therefore D_B = Q_B^+$

$\Rightarrow Q_A^+ = D_A = x Q_A + Q_B$

$\Rightarrow Q_B^+ = \bar{Q}_A Q_B$

$\Rightarrow y = \bar{x}\bar{Q}_B + x Q_A$

Step 3: state diagram



Note: If 2 JK FF are given, find J_A, K_A, J_B & K_B using the circuit

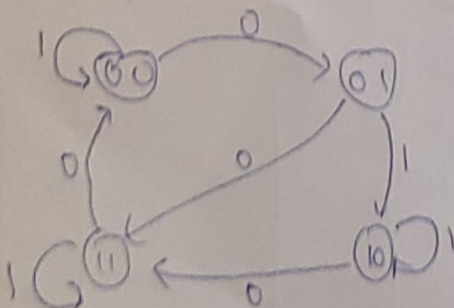
ex: $J_A = Q_B$ $K_A = \bar{x} J_A$

$J_B = \bar{x}$ $K_B = x \oplus Q_B$

State table:

P.S Q_A	P.S Q_B	i/p x	J_A	K_A	J_B	K_B	N.S Q_A^+	N.S Q_B^+
0	0	0	0	0	1	0	0	1
0	0	1	0	0	1	0	0	0
0	1	0	1	1	1	1	1	1
0	1	1	1	1	1	1	1	0
1	0	0	0	0	1	0	1	1
1	0	1	0	0	1	0	1	0
1	1	0	1	1	0	1	0	0
1	1	1	1	0	0	0	1	1

- Fill J_A & K_A, J_B & K_B using eqn found.
- Use TT of JK Flip Flop to fill Q_A^+ & Q_B^+



$Q_A Q_B$

$S_0 = 00$

$S_1 = 01$

$S_2 = 10$

$S_3 = 11$